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Abstract

The goal of this project is to design a low-cost control circuit for a solar-powered battery charger.

To reduce system cost and complexity, a combination of analog and digital logic circuits are used to generate control signals. The completed system demonstrates the feasibility of a low-cost, microcontroller-free solar charging system capable of regulating output voltage. The project highlights the effectiveness of analog control techniques in renewable energy applications where simplicity, reliability, and low power consumption are desired.

Introduction

Solar power is incredibly versatile and **becoming more** accessible as the price of solar cells continue to drop. In cases where connection to the grid is not possible—in emergencies or rural areas—solar is a convenient and easy solution, the only requirement being sunlight.

For my capstone project, I **designed** a low-power solar charger for a rechargeable Li-ion power bank. A buck converter **steps** down the voltage output of a 25W solar panel to a level suitable for charging the power bank. To reduce complexity **and cost**, I decided against using a microcontroller for the control system.

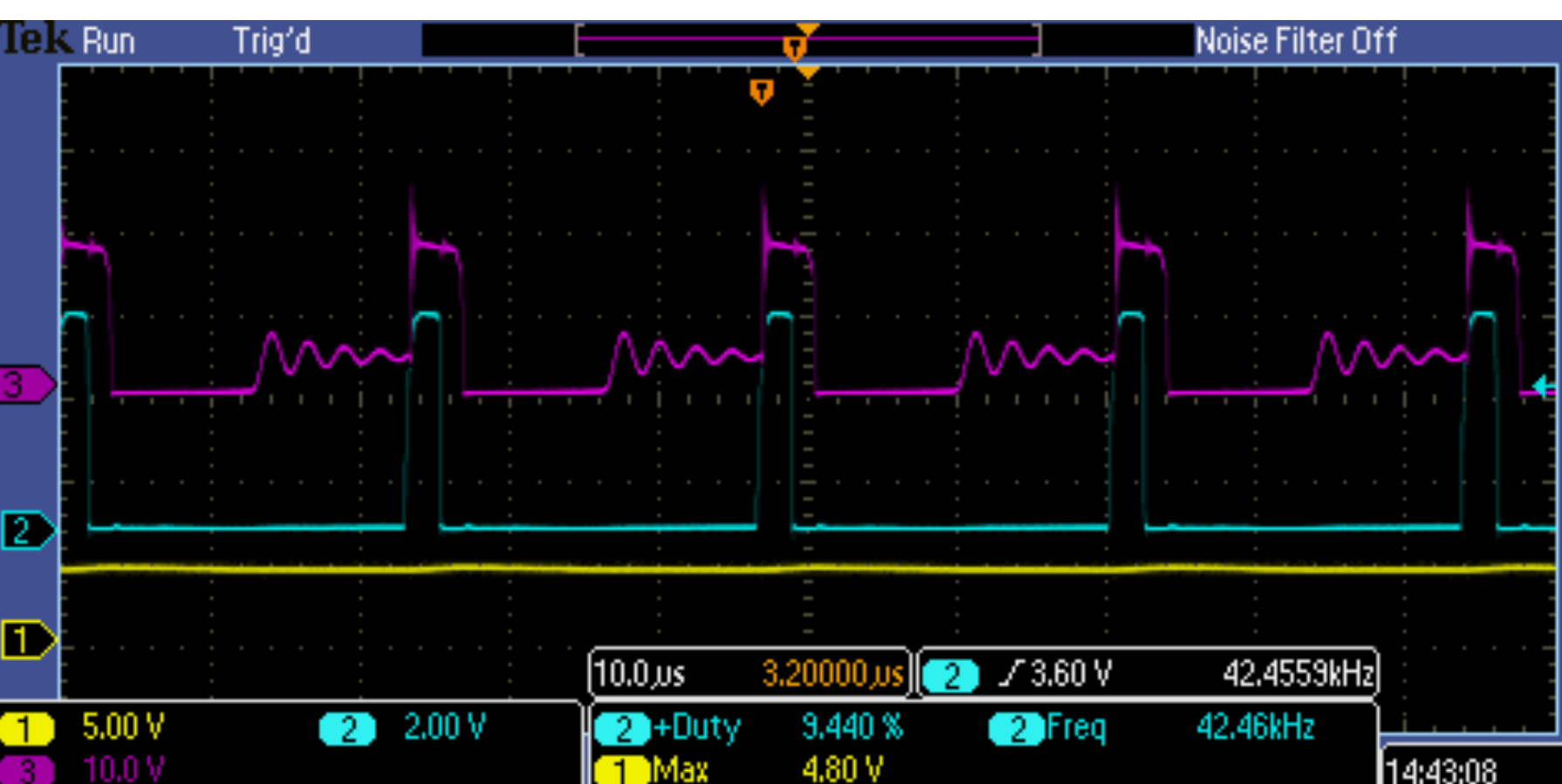


Image of oscilloscope waveforms when the input voltage is 17V. The purple waveform shows the voltage across the diode, where ringing is visible. The blue waveform shows the duty cycle, and the output voltage is shown in yellow.

Approach

Rather than a microcontroller, digital logic circuits were used to generate control signals:

- Difference between input and 5V reference
- Difference is scaled down to a range of 0-3.5V
- Summed with 1.5V bias for an error signal between 1.5-5V **These three bullets are not clear individually. It seems you're trying to describe a series of steps taken to generate the error signal. In that case, combine the three bullets into one to make it more clear.**
- An LM555 timer circuit is used to generate a sawtooth signal, which is compared with the error signal to produce a variable PWM signal for the

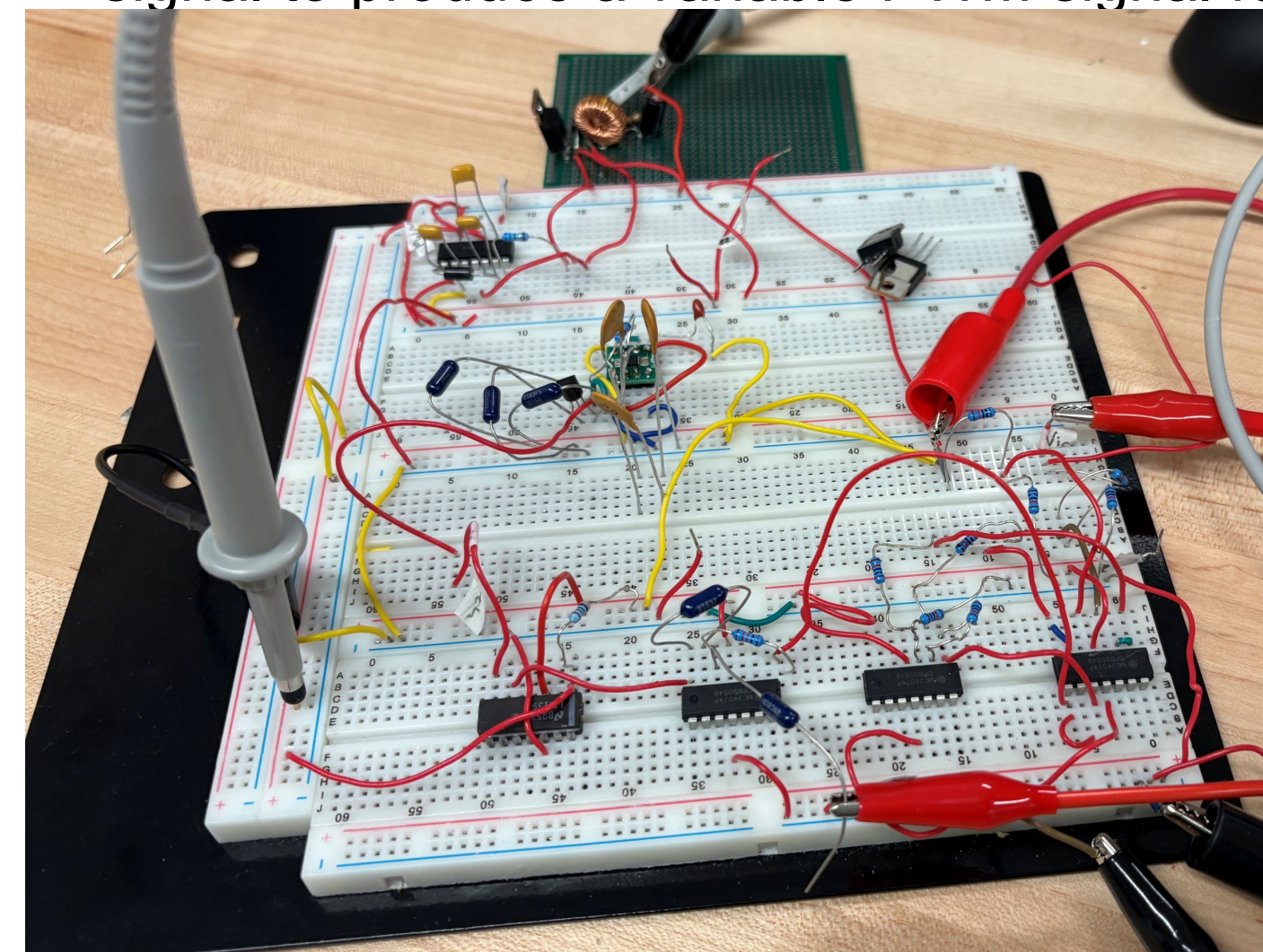
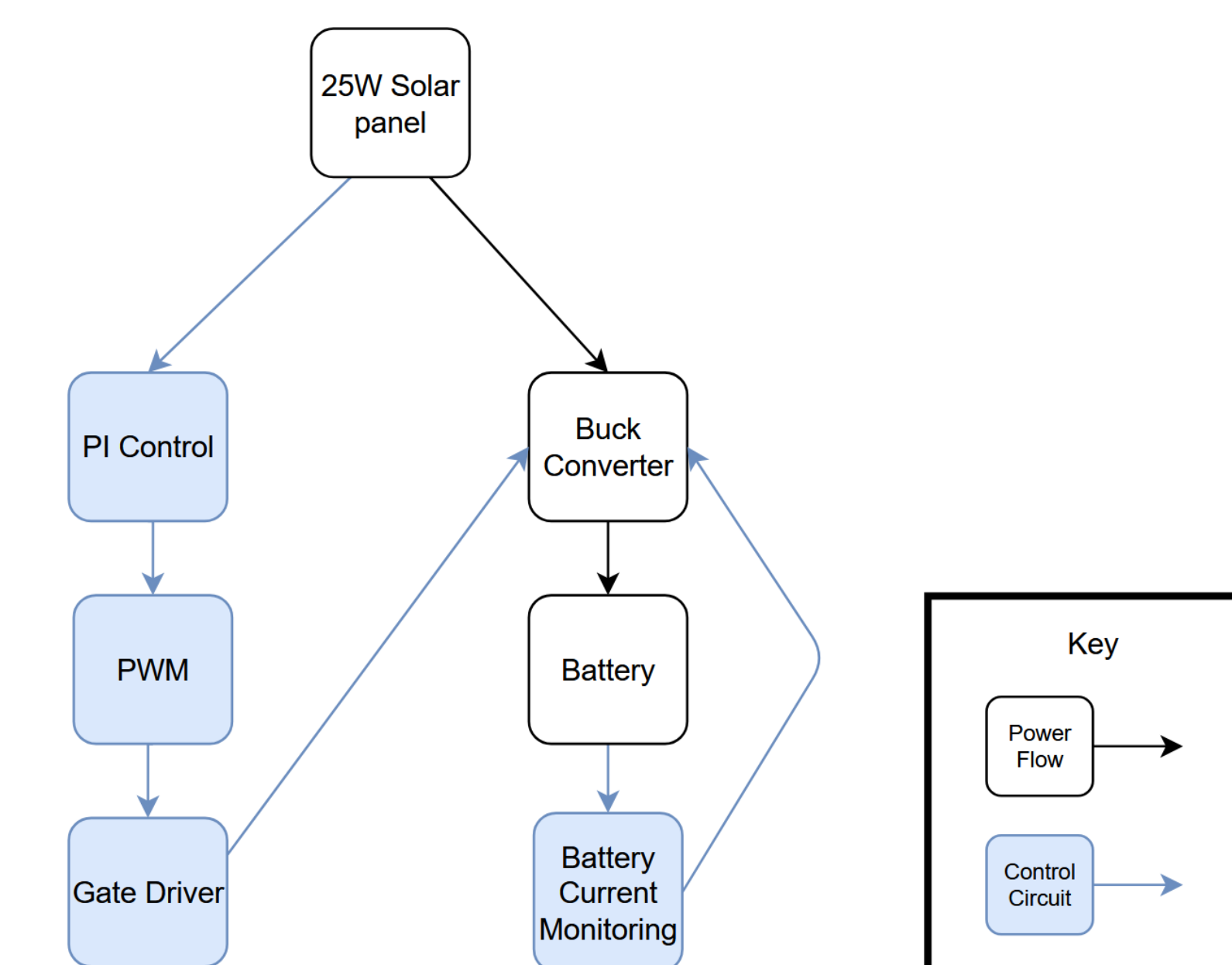


Image of control circuit (on breadboard) and buck converter (on perfboard).

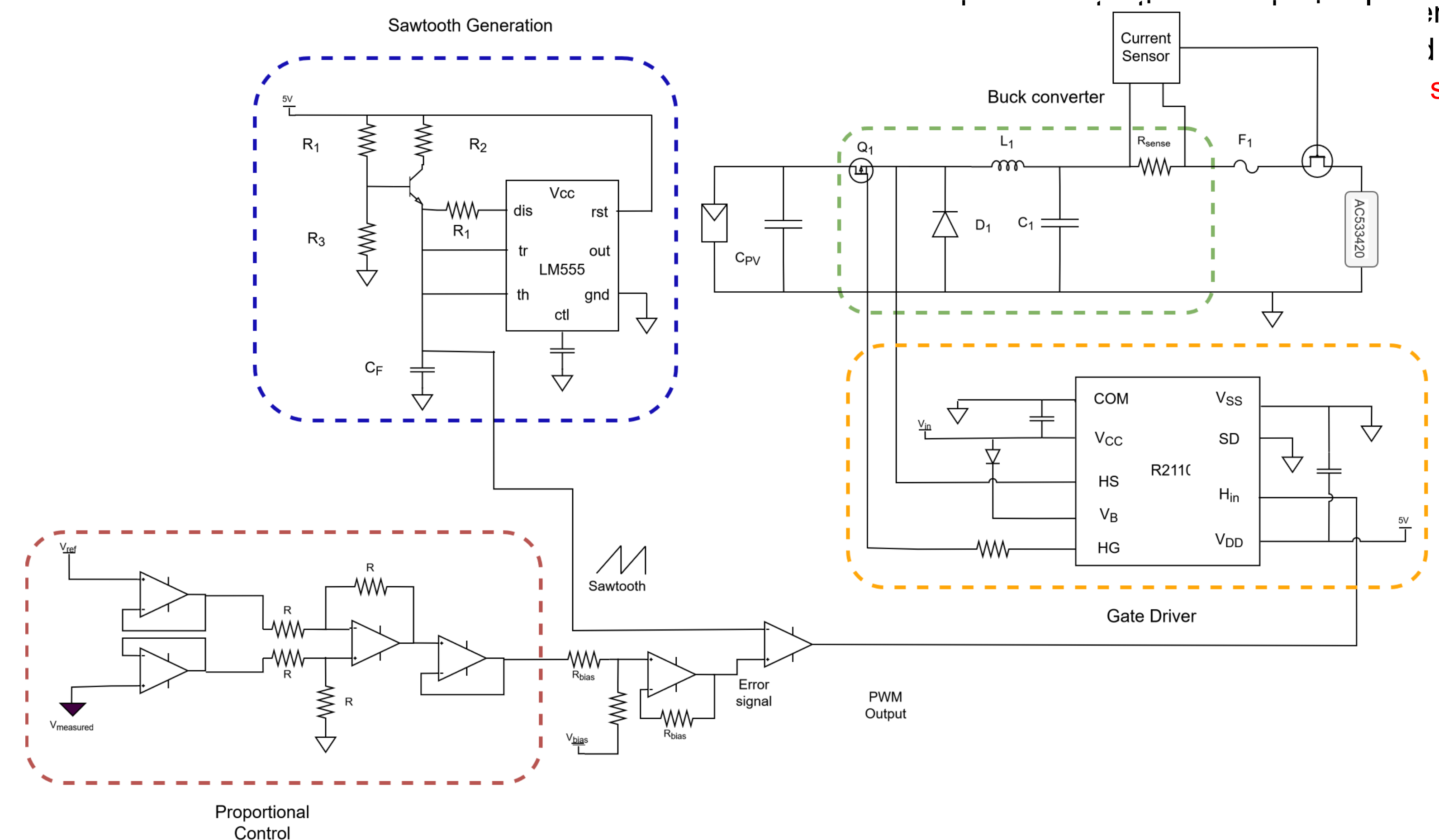


Results

The system successfully steps down the PV voltage to an acceptable level for charging a battery. The control circuit is responsive to changes to the input voltage and accurately alters the duty cycle of the PWM to reflect these changes.

Excluding the cost of the solar panel and the battery, the total cost of parts is around \$10. this fulfills the requirement that the system is low cost.

Unfortunately, the implementation of the buck converter on a perfboard rather than a PCB introduces enough parasitic capacitance and inductance—appearing as ringing across the diode—to noticeably increase the output of the buck converter as the input voltage decreases, though this will not affect the battery's ability to charge.



Block diagram of the system. Black denotes power flow, blue denotes control signals.

Future Work & Improvements

In this implementation, the maximum power point voltage is held constant. The original plan for this project included a simple switch to swap between two different values of V_{MPP} , **but this was cut for time. You don't need to describe the original plan and why it was not implemented. Just state what you will implement if you had time.** This is the most obvious improvement that can be made, as it would increase power efficiency while keeping costs low. Both logic circuits or MPPT ICs can be used for this.

Additionally, implementation of the buck stage on a PCB would have further reduced if not eliminated the parasitics.

Finally, to support a wider variety of load batteries, **implemented, such as 1 voltage circuit diagram of the system.**